

Supplementary Materials to “Learning and Encoding Temporally Structured Sensory Inputs via Superburst Dynamics”

Inhoi Jeong

Department of Physics, Korea University, Seoul 02841, Korea

Jongmu Kim

Department of Mechanical Engineering, Korea University, Seoul 02841, Korea

Kyoung J. Lee[†]

Department of Physics, Korea University, Seoul 02841, Korea

[†] Corresponding author (email: kyoung@korea.ac.kr)

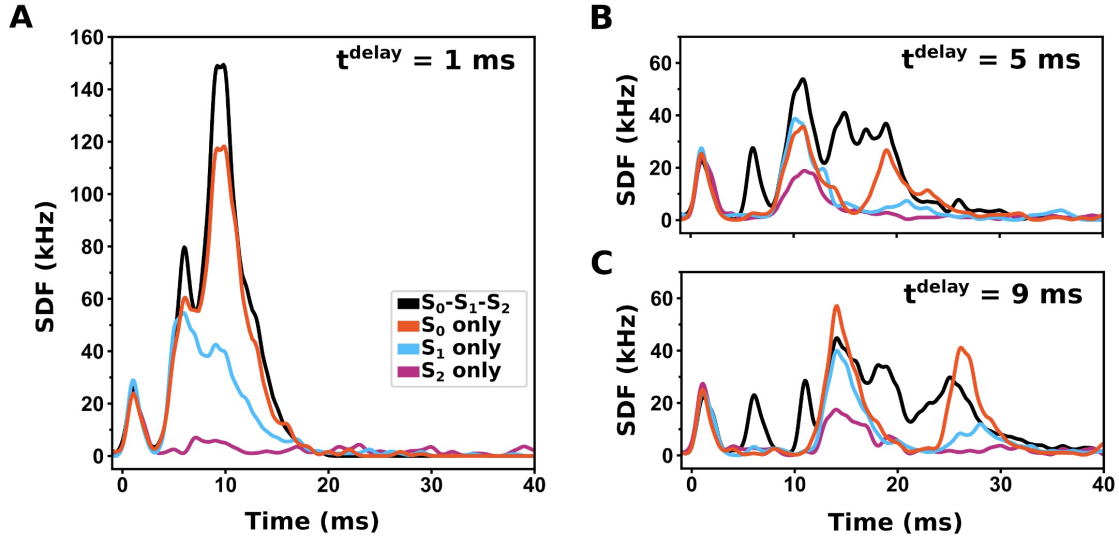


Figure S1. Spike density functions in response to different stimuli presented to three networks conditioned with different values of conduction time delay t^{delay} . The test triplet stimulus $S_0-S_1-S_2$ has time intervals of $\Delta t_1 = \Delta t_2 = 5 \text{ ms}$. “Time” denotes time from stimulus onset.

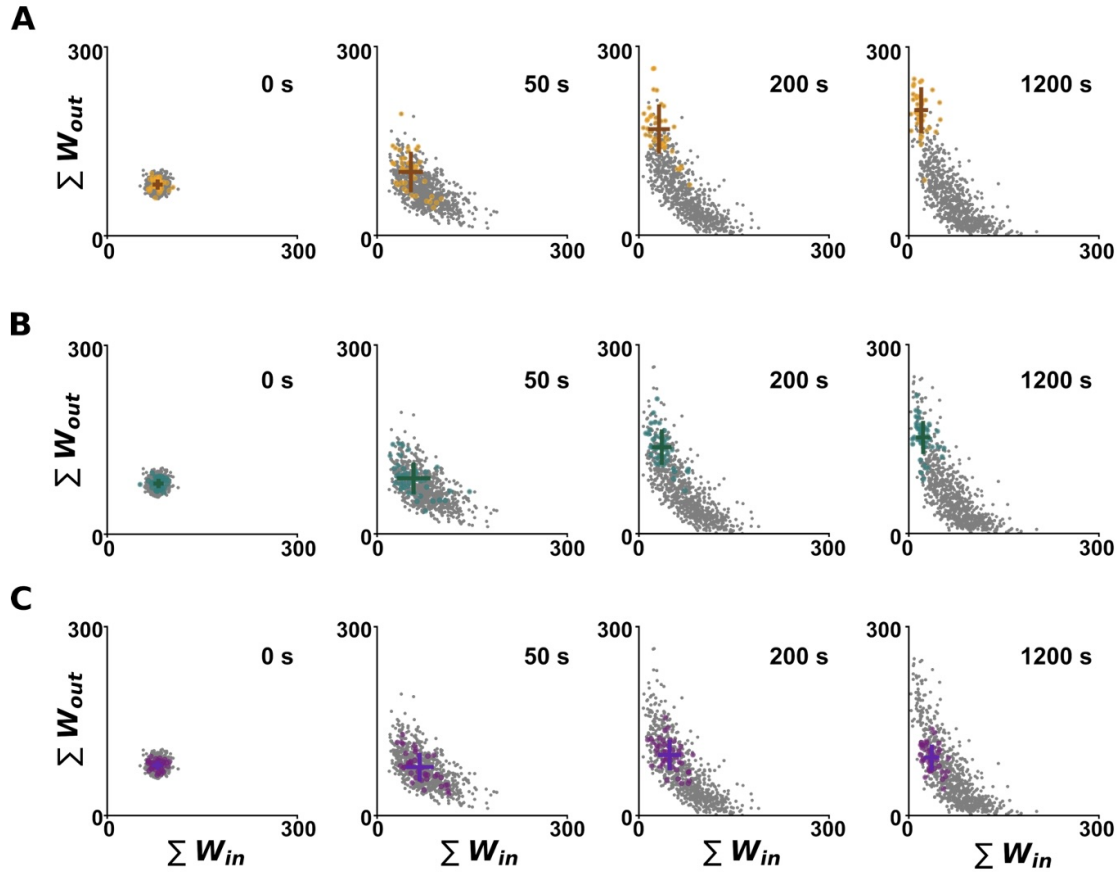


Figure S2. Time evolution of the mean positions of S_0 (yellow), S_1 (green) and S_2 (violet) subject to the Pavlovian conditioning: See Fig. 3B in the main text.

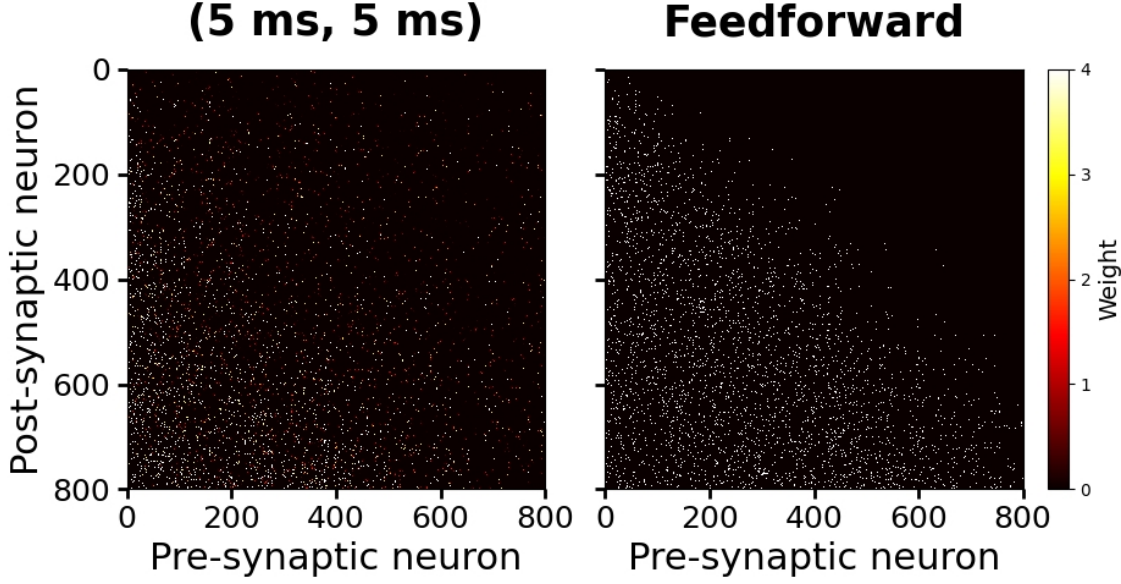


Figure S3. Synaptic weight 2D matrix for a conditioned network with $\Delta t_1^{\text{cond.}} = \Delta t_2^{\text{cond.}} = 5$ ms (LEFT) and an artificially constructed, purely feedforward network (RIGHT). The 2D matrix shown in the left is essentially an alternative representation of the 3D weight distribution shown in Fig. 3C ($t = 1200$ s). The weight matrix in the right is generated as follows: (1) 800 excitatory neurons are indexed from 1 (top) to 800 (bottom); (2) assuming a sparsity of 0.1, a total of $0.1 \times (0.5 \times 800 \times 799)$ downward connections are randomly assigned; and (3) neuron indices are reordered according to their $\sum W_{\text{in}}$.

Decoding network training

Input-triggered SBs in the conditioned network form stimulus-specific population signatures that can be classified by a downstream spiking decoder. To classify different SB firing patterns, we trained networks of 512 or 256 ALIF neurons using the e-prop (eligibility propagation) algorithm [35]. Our implementation customizes the previously published “tonic classifier” code, originally developed for recognizing the Spiking Heidelberg Digits (SHD) dataset [36].

E-prop factorizes the gradient of the loss with respect to each synaptic weight into the product of a local eligibility trace and a neuron-specific learning signal. For ALIF neurons, eligibility traces incorporate both membrane potential and adaptation dynamics, enabling biologically plausible credit assignment without backpropagation through time. Weight updates follow

$$\Delta W_{ij} \propto \sum_t e_{ij}(t) L_j(t),$$

where $e_{ij}(t)$ is the eligibility trace for synapse $i \rightarrow j$, and $L_j(t)$ is the learning signal derived from the output error.

Neurons in the recurrent layer are connected in an all-to-all fashion to output layer nodes, each dedicated to classifying one of ten distinct SB patterns. Both recurrent

synapses and output weights were trained using supervised e-prop learning. Each training epoch was based on a batch size of 512. The number of data used for the training (test) session is 67,362 (16,841).

Raster plots representing the activity of 800 excitatory encoder neurons for 10 distinct stimulus types [the 9 cases shown in Fig. 5A together with the non-target case (S_4, S_5, S_6) from Fig. 6C] are fed into a recurrent spiking neural network composed of either 256 or 512 adaptive leaky integrate-and-fire (ALIF) neurons. The recurrent network projects to 10 output channels, each corresponding to one SB class. Recurrent and output synaptic weights are trained using the supervised e-prop algorithm. Increasing the recurrent-layer size from 256 to 512 neurons yields a modest improvement in test accuracy, from $92.5 \pm 3.2\%$ to $94.7 \pm 1.2\%$, while both models achieve $\sim 99.5\%$ training accuracy (see Fig. S5). Classification performance also depends strongly on synaptic connectivity sparsity. The two configurations shown correspond to near-optimal performance obtained from systematic variation of network size and sparsity parameters.

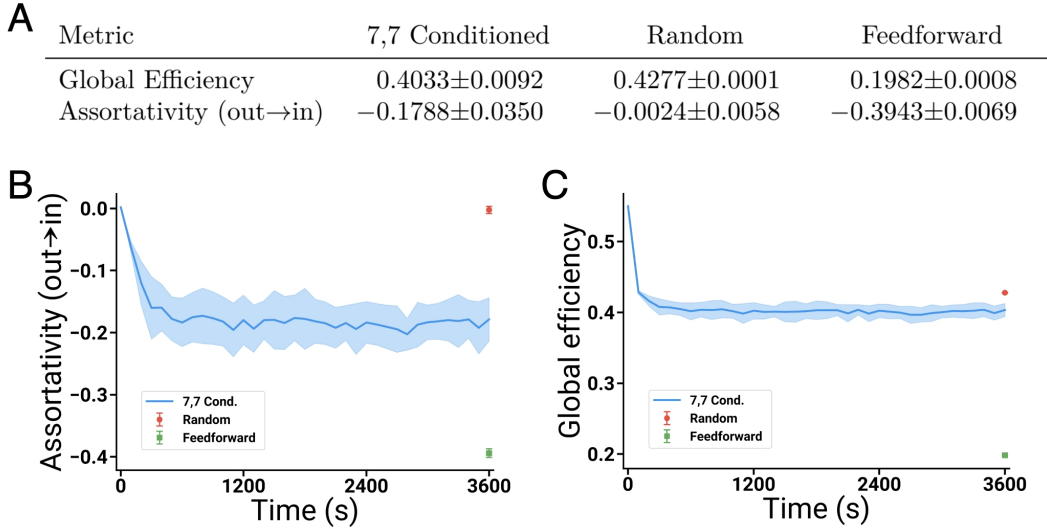


Figure S4. Comparison of two graph-theoretic measures for three different types of networks. (A) Table summarizing the global efficiency and (out $\rightarrow$ in) assortativity for a Pavlovian conditioned network (with $\Delta t_1^{\text{cond.}} = \Delta t_2^{\text{cond.}} = 7$ ms), a random network, and a purely feedforward network without recurrent feedback connections. (B) and (C) show the temporal evolution of the two measures (averaged over 10 independent runs) during the conditioning process. The steady-state values (measured at $t = 3600$ s) are summarized in the table shown in (A).

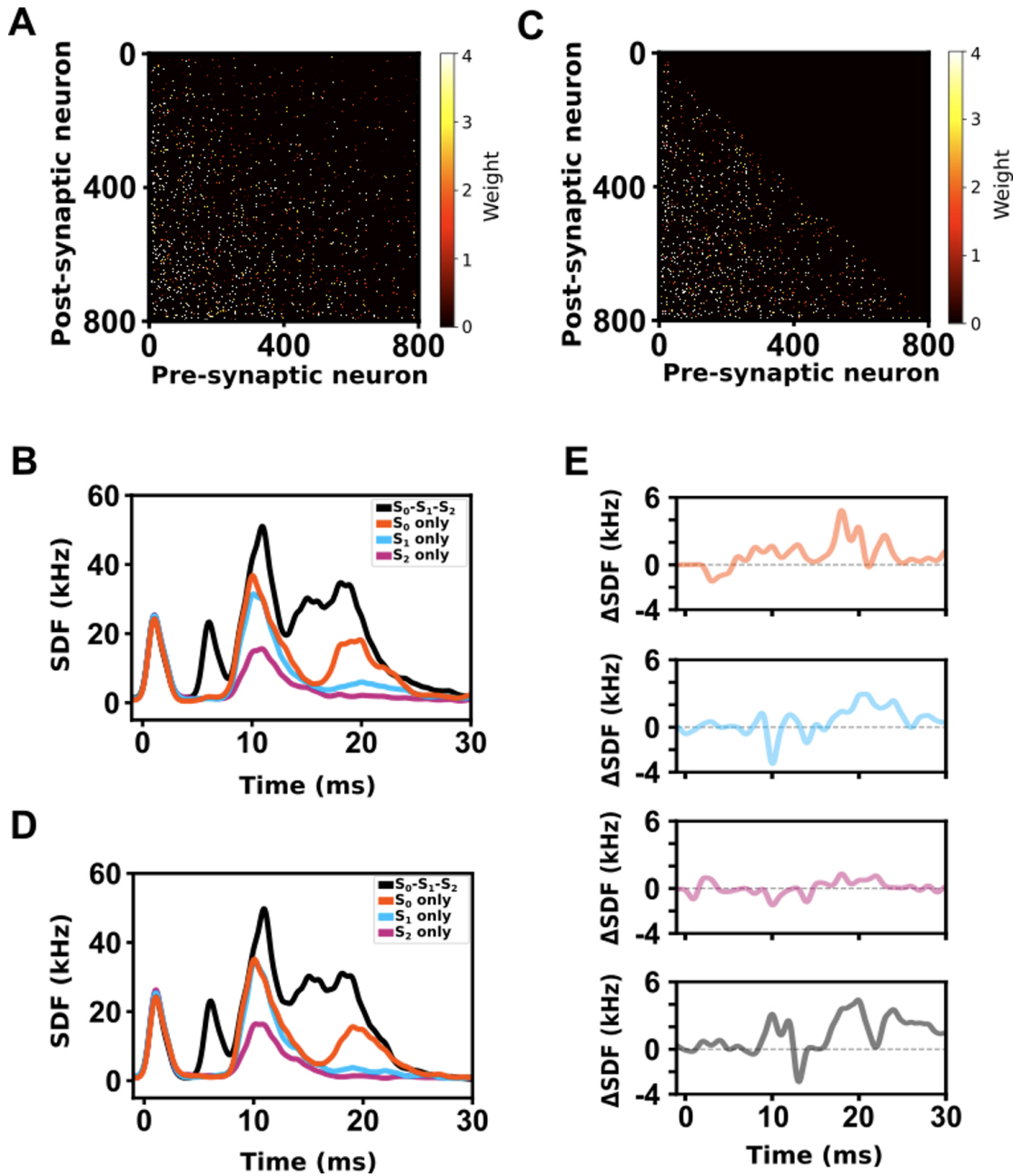


Figure S5. The impact of removing all feedback connections (A) depicts the weight matrix of the conditioned network (see Fig. 3, $t = 1200$), while (C) depicts (A) with all feedback connections removed. (B) and (D) are responses to different stimuli for the case of (A) and (C), respectively. (E) are the differences (color-matched) traces between (B) and (D). “Time” represents the time from stimulus onset.

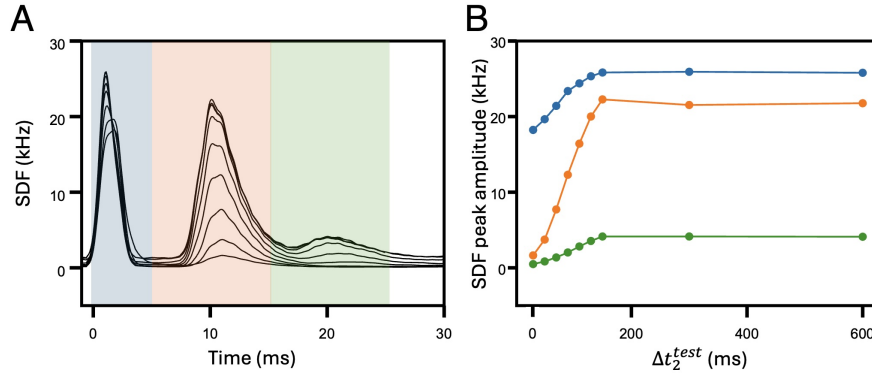


Figure S6. Estimation of SB refractory time (A) SDF activation profiles immediately following the S_2 stimulation (of $S_0 - S_1 - S_2$ triplet) for different values of Δt_2^{test} ($\Delta t_1^{\text{test}} = 5$ ms was kept for all cases.) The network was conditioned with $\Delta t_1^{\text{cond.}} = \Delta t_2^{\text{cond.}} = 5$ ms, prior to acquiring these profiles. (B) Plots of three local SDF maxima of the SDF profiles shown in (A) as a function of Δt_2^{test} . Note that all three peak values approach to their saturation around $\Delta t_2^{\text{test}} = 170$ ms.

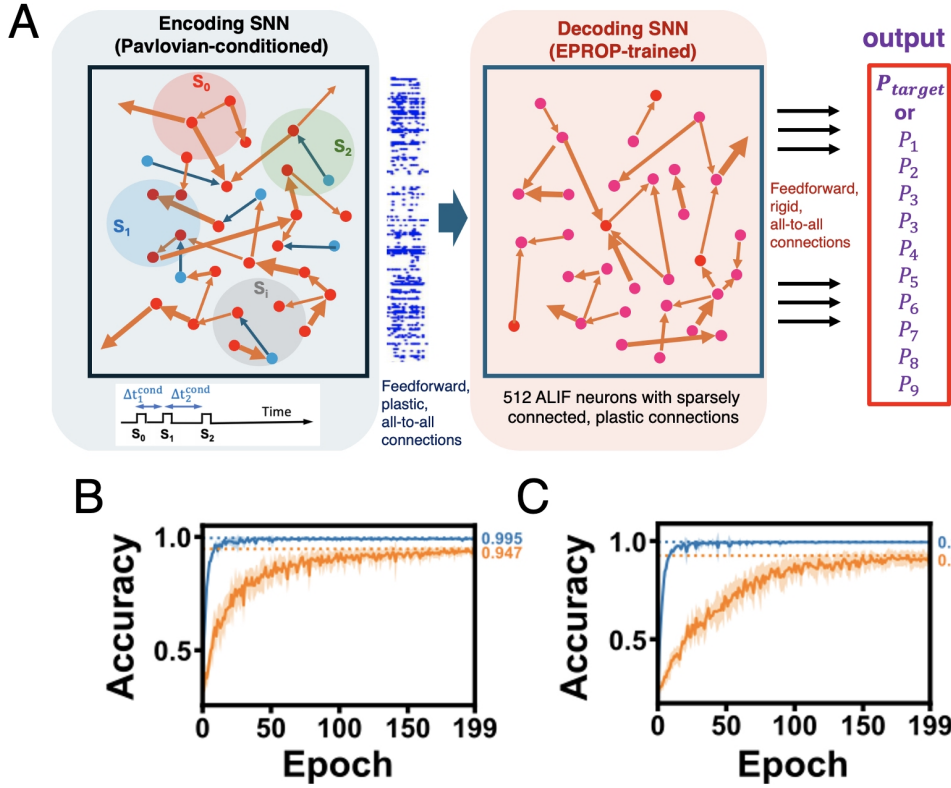


Figure S7. Sorting accuracy of decoding network as a function of epoch. (A) Schematic illustration of a decoding network (and a classifier) receiving the output SB signal from a Pavlovian conditioned encoder. In (B), the decoder contains 256 ALIF neurons with sparsities of 0.6 (encoder-to-recurrent) and 1.0 (recurrent-to-recurrent). In (C), the decoder contains 512 ALIF neurons with corresponding sparsities of 1.0 and 0.2. Blue and orange traces denote training and test accuracy, respectively, averaged over 10 independent runs; shaded regions indicate $\pm$ standard deviation. “Time” represents the time from stimulus onset.